classify_to_functional_type.R

August 10, 2026

Contents

classify_to_functional_type 1

Index **3**

`classify_to_functional_type` *Classify Community Data to Functional Types*

Description

Classifies taxa in a long-format community data frame to functional-type (FT) labels using a precomputed FT classification table. Aggregates pollen proportions by all non-'taxon'/'value' identifier columns plus functional type. Produces the same output column structure as 'classify_taxonomic_resolution()' so it can be used as a drop-in replacement downstream.

Usage

```
classify_to_functional_type(
  data_source,
  data_functional_type_classification,
  verbose = TRUE
)
```

Arguments

`data_source` A data frame containing community data with columns 'taxon', 'dataset_name', 'age', and 'value'. Other identifier columns such as 'sample_name' are preserved.

`data_functional_type_classification` A data frame mapping taxa to functional types. Must contain columns 'taxon_name' (character) and 'functional_type' (integer). Typically produced by 'assign_functional_type_clusters()' and loaded via 'load_latest_functional_type_classification()'.

`verbose` Logical. Reserved for informational progress messages. The warning emitted when unmatched taxa are dropped is always shown because it reports data loss.

Details

Steps performed:

1. Validate arguments.
2. Left-join ‘data_source‘ to ‘data_functional_type_classification‘ on their taxon names.
3. Drop unmatched taxa (NA functional type) with a warning.
4. Create ‘taxon‘ labels ‘"FT_functional_type"‘.
5. Aggregate ‘value‘ by original identifier columns and ‘taxon‘.
6. Full-join back to an identifier/taxon cross-reference to preserve true negatives.

Value

A data frame with the same column names as ‘data_source‘. The ‘taxon‘ column is replaced by functional-type labels of the form ‘"FT_1"‘, ‘"FT_2"‘, etc. ‘value‘ is aggregated (summed) by all original identifier columns and ‘taxon‘. All identifier combinations present after FT classification are preserved (true negatives kept via a cross-reference join). Taxa not found in ‘data_functional_type_classification‘ are dropped with a ‘cli::cli_warn()‘ message.

See Also

[classify_taxonomic_resolution()], [load_latest_functional_type_classification()], [assign_functional_type_clusters()]

Index

classify_to_functional_type, 1